Agile Scrum

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One of the largest and fastest-growing Agile methodologies, Scrum is a simple project management framework that has a small set of interrelated practices and principles, is not overly prescriptive, and is able to produce productivity gains for software development teams almost immediately.

Grameen Solutions Ltd.

Contents

Audience

This hands-on training course provides an excellent introduction to the Scrum Framework, and as such it is appropriate for anyone implementing or working alongside a timebox-based Agile team.

In addition to ScrumMasters, Product Owners, Developers and Testers, other roles that would benefit from this course include Product Managers and Analysts, Project Managers, Team Leads, Architects, Functional Managers, CIOs, and CTOs.

Learning Objectives

After this session, attendees will be able to:

- Describe how the Scrum Framework is based on empirical process and Lean, and how Scrum relates to the values and principles of the Agile Manifesto.
- Identify the three Scrum roles and their responsibilities and describe why these roles form the Scrum Team; explore impact on other roles.
- Understand how the ScrumMaster protects, guides, and serves the team while also acting as a change agent.

- Identify and know how to effectively facilitate Scrum ceremonies: Sprint Planning, Daily Scrum, and Sprint Demo/ Review/Retrospective, Release Planning.
- Identify and understand the value of Scrum Artifacts: Product Backlog, Product Increment and Definition of Done, Sprint Backlog, Burndown Charts.
- Understand how to lead teams effectively in a large-scale environment. Know what roles and artifacts are added.
- Identify common organizational obstacles to Scrum and Agile adoption and understand how the ScrumMaster can help navigate those obstacles.

Overview:

Agile is an umbrella term used to describe a general approach to software development. Though there are many agile incarnations, all agile processes, including **Scrum** emphasize teamwork, frequent deliveries of working software, close customer collaboration, and the ability to respond quickly to change.

What is Agile

Agile is one of the big buzzwords of the IT development industry. But exactly what is agile development? Put simply, agile development is a different way of managing IT development teams and projects. The use of the word agile in this context derives from the agile manifesto.

1. Individuals and interactions over processes and tools

- in agile development, self-organization and motivation are important, as are interactions like co-location and pair programming.

2. Working software over comprehensive documentation
   - working software will be more useful and welcome than just presenting documents to clients in meetings.
3. Customer collaboration over contract negotiation
   - requirements cannot be fully collected at the beginning of the software development cycle, therefore continuous customer or stakeholder involvement is very important.
4. Responding to change over following a plan
   - agile development is focused on quick responses to change and continuous development

Agile principles

The Agile Manifesto is based on twelve principles:

1. Customer satisfaction by rapid delivery of useful software
2. Welcome changing requirements, even late in development
3. Working software is delivered frequently (weeks rather than months)
4. Close, daily cooperation between business people and developers
5. Projects are built around motivated individuals, who should be trusted
6. Face-to-face conversation is the best form of communication (co-location)
7. Working software is the principal measure of progress
8. Sustainable development, able to maintain a constant pace
9. Continuous attention to technical excellence and good design
10. Simplicity—the art of maximizing the amount of work not done—is essential
11. Self-organizing teams
12. Regular adaptation to changing circumstances

Iterative vs. Waterfall

1) One of the differences between agile and waterfall is that testing of the software is conducted at different stages during the software development lifecycle. In the Waterfall model, there is always a separate testing phase near the completion of an implementation

phase. However, in Agile and especially Extreme programming, testing is usually done concurrently with coding, or at least, testing jobs start in early iterations.

Code vs. Documentation

2) "Working software over comprehensive documentation" as "We want to spend all our time coding. Remember, real programmers don't write documentation."

Agile methods

Well-known agile software development methods and/or process frameworks include:

1. Adaptive Software Development (ASD)
2. Agile Modeling
3. Agile Unified Process (AUP)
4. Crystal Methods (Crystal Clear)
5. Disciplined Agile Delivery
6. Dynamic Systems Development Method (DSDM)
7. Extreme Programming (XP)
8. Feature Driven Development (FDD)
9. Lean software development
10. Kanban
11. Scrum
12. Scrum-ban

What is Scrum

Scrum is the leading agile development methodology, used by Fortune 500 companies around the world. The Scrum Alliance exists to transform the way we tackle complex projects, bringing the Scrum framework and agile principles beyond software development to the broader world of work.

a) Scrum is an agile way to manage a project, usually software development
b) left up to the Scrum software development team. This is because the team will know best how to solve the problem they are presented.
c) Scrum relies on a self-organizing, cross-functional team
d) Within agile development, Scrum teams are supported by two specific roles.

i) The first is a ScrumMaster, who can be thought of as a coach for the team, helping team members use the Scrum process to perform at the highest level.
ii) The product owner (PO) is the other role, and in Scrum software development, represents the business, customers or users, and guides the team toward building the right product.

Why Scrum

a. Because you don't want to waste your time and money building a product no one will want to use or pay for. This is where MVP (minimum viable product) comes in; creating a product with the minimum set of product features that will give value to users, in order to test the concept and get feedback before continuing development.

b. The Scrum processes are very clear about the need to produce visible value in the form of working software on a regular basis. By delivering a product iteratively and incrementally (i.e. in sprints), we also maximize the opportunity for regular customer feedback and the return on investment.

c. 

Key benefits of using Scrum

Based on the industry standard of Scrum - A Guide to Scrum Body of Knowledge (SBOK™ Guide), some of the key benefits of using Scrum in any project are:

1) **Adaptability** - Empirical process control and iterative delivery make projects adaptable and open to incorporating change.
2) **Transparency** - All information radiators like a Scrumboard and Sprint Burndown Chart are shared, leading to an open work environment.
3) **Continuous Feedback** - Continuous feedback is provided through the Conduct Daily Standup, Demonstrate and Validate Sprint processes.

4) **Continuous Improvement** - The deliverables are improved progressively Sprint by Sprint, through the Groom Prioritized Product Backlog process.
5) **Continuous Delivery of Value** - Iterative processes enable the continuous delivery of value through the Ship Deliverables process as frequently as the customer requires.
6) **Sustainable Pace** - Scrum processes are designed such that the people involved can work at a sustainable pace that they can, in theory, continue indefinitely.
7) **Early Delivery of High Value** - The Create Prioritized Product Backlog process ensures that the highest value requirements of the customer are satisfied first.
8) **Efficient Development Process** - Time-boxing and minimizing non-essential work leads to higher efficiency levels.
9) **Motivation** - The Conduct Daily Standup and Retrospect Sprint processes lead to greater levels of motivation among employees.
10) **Faster Problem Resolution** - Collaboration and colocation of cross-functional teams lead to faster problem solving.
11) **Effective Deliverables** - The Create Prioritized Product Backlog process and regular reviews after creating deliverables ensures effective deliverables to the customer.
12) **Customer Centric** - Emphasis on business value and having a collaborative approach to stakeholders ensures a customer-oriented framework.
13) **High Trust Environment** - Conduct Daily Standup and Retrospect Sprint processes promote transparency and collaboration, leading to a high trust work environment ensuring low friction among employees.
14) **Collective Ownership** - The Approve, Estimate, and Commit User Stories process allows team members to take ownership of the project and their work leading to better quality.
15) **High Velocity** - A collaborative framework enables highly skilled cross-functional teams to achieve their full potential and high velocity.
16) **Innovative Environment** - The Retrospect Sprint and Retrospect Project processes create an environment of introspection, learning, and adaptability leading to an innovative and creative work environment.

Scrum vs. Traditional Project Management

1. Traditional project management emphasizes on conducting detailed upfront planning for the project with emphasis on fixing the scope, cost and schedule - and managing those parameters.
2. Whereas, Scrum encourages data-based, iterative decision making in which the primary focus is on delivering products that satisfy customer requirements.

3. To deliver the greatest amount of value in the shortest amount of time, Scrum promotes prioritization and Time-boxing over fixing the scope, cost and schedule of a project.
4. An important feature of Scrum is self-organization, which allows the individuals who are actually doing the work to estimate and take ownership of tasks.

Following table summarizes many of the differences between Scrum and traditional project management:

Parameters	Scrum	Traditional Project Management
Emphasis is on	People	Processes
Documentation	**Minimal - only as required**	**Comprehensive**
Process style	**Iterative**	**Linear**
Upfront planning	Low	High
Prioritization of Requirements	**Based on business value and regularly updated**	**Fixed in the Project Plan**
Quality assurance	**Customer centric**	**Process centric**
Organization	**Self-organized**	**Managed**
Management style	Decentralized	Centralized

Change	Updates to Productized Product Backlog	Formal Change Management System
Leadership	**Collaborative, Servant Leadership**	**Command and control**
Performance measurement	Business value	Plan conformity
Return on Investment	**Early/throughout project life**	**End of project life**
Customer involvement	High throughout the project	Varies depending on the project lifecycle

Scrum Roles

Product Owner

- The Scrum product owner is typically a project's key stakeholder.
- Maximize business value by the Team
- Maintain and prioritize the Product Backlog sequentially (1 to n)
- Create and maintain the Release Burndown Chart
- Help the ScrumMaster organize Sprint Review Meetings
- Attend Scrum Meetings
- Clearly communicate the business case to the Team and Stakeholders
- Build and maintain a relationship with the Stakeholders
- Support the ScrumMaster to help the Team become self-organizing
- Report progress to the Stakeholders regularly

- Ensure the proper use of corporate resources and assist the Team to obtain resources as needed

Scrum Master

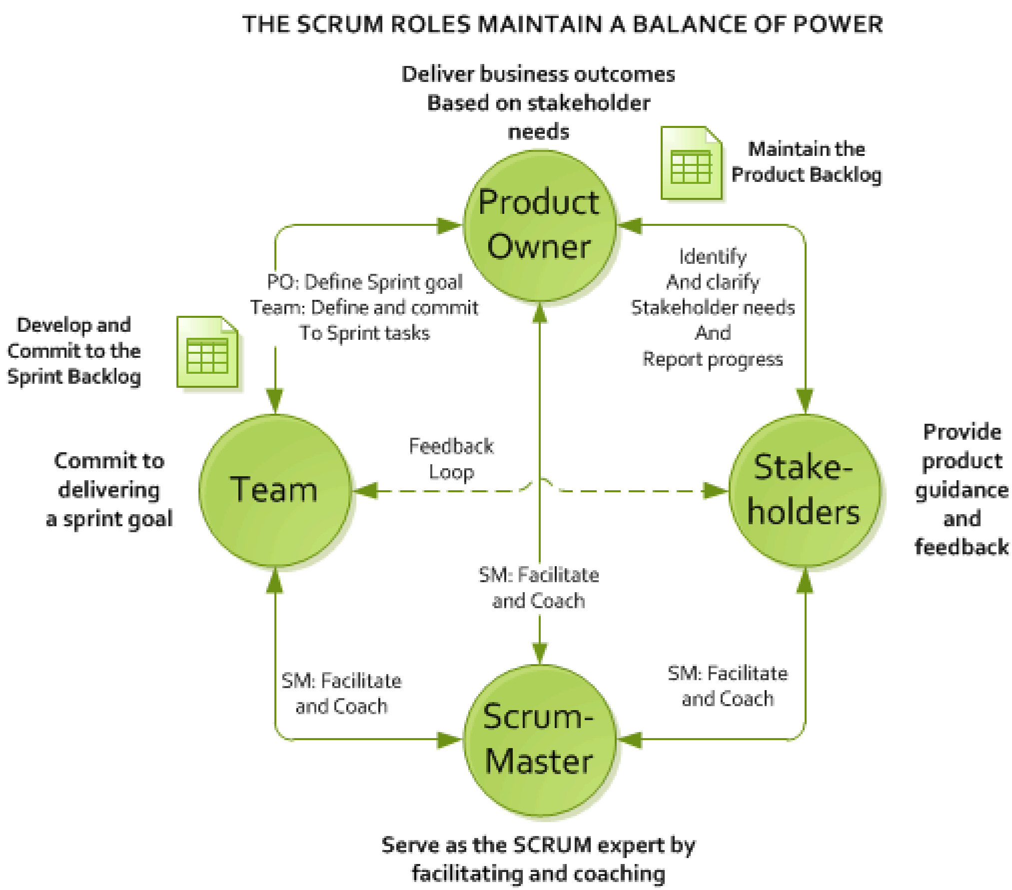
- Communicate the value of Scrum
- Teach the organization on Scrum to maximize business value
- Facilitate Sprint Planning, Daily Scrums, Sprint Reviews and Retrospective Meetings
- Create the Task Board and Sprint Burndown Chart at the start of every Sprint
- Attend all Scrum meetings
- Preserve the integrity and spirit of the Scrum framework
- Maintain the focus of the Team
- Make the Team aware of impediments and facilitate efforts to resolve them
- Serve as a coach and mentor to members of the Team
- Respectfully hold the Team, Product Owner and Stakeholders accountable for their commitments
- Continually work with the Team and business to find and implement improvements

Team

- Typically 5-9 people
- Cross-functional:
  - Programmers, testers, user experience designers, etc.
- Members should be full-time
  - May be exceptions (e.g., database administrator)

Stakeholders

- Work with the Product Owner to develop and maintain the Product Backlog
- Attend Sprint Planning meetings as needed to provide feedback and expertise
- Provide direct feedback to the Team during Sprint Reviews
- Remove roadblocks and impediments for the Team, Product Owner and ScrumMaster
- Avoid distracting the Team during a Sprint — after the Team has committed to the Sprint
- Support the Scrum Framework



Scrum Meetings or Ceremonies

Release Planning/Backlog grooming

1. Agile release planning is best accomplished via the straightforward method of getting everyone into a room together.
2. Everybody who will be involved in the release should participate in the planning. That gives us a potential invite list that looks like this: Scrum Master, Product Owner, Delivery Team, Stakeholders, Outside experts, Customer(s), 3rd Party Vendors, Marketing, Sales
3. A Release Planning agenda might look like this:
    1) Goals of the release (PO)
    2) Background, business and competitive climate (PO)
    3) Current product and development state (PO)
    4) Grooming of Product Backlog for release (All)
    5) Capture and discussion of issues (All)
    6) Technical Issues (Dev teams)
        a) Technology
        b) Testing Challenges and Strategies
        c) Dependencies
        d) Engineering Standards and Practices
        e) Hardening and Hackathon
        f) Development regimen
            i) Build
            ii) Test
            iii) Continuous integration
    7) Team interactions (Dev teams)
        a) Scrum of Scrums
        b) Special handling of multi-team Sprint Planning, Sprint Reviews
        c) Multi-team retrospectives
    8) Tentative schedule (PO)

Sprint Planning

1. In Scrum, the sprint planning meeting is attended by the product owner, ScrumMaster and the entire Scrum team. Outside stakeholders may attend by invitation of the team, although this is rare in most companies.
2. During the sprint planning meeting, the product owner describes the highest priority features to the team. The team asks enough questions that they can turn a high-level user story of the product backlog into the more detailed tasks of the sprint backlog.
3. The product owner doesn't have to describe every item being tracked on the product backlog. A good guideline is for the product owner to come to the sprint planning meeting prepared to talk about two sprint's worth of product backlog items.
4. Analyze and evaluate product backlog
5. There are two defined artifacts that result from a sprint planning meeting:
    a. A sprint goal
        1. A sprint goal is a short, one- or two-sentence, description of what the team plans to achieve during the sprint.
        2. It is written collaboratively by the team and the product owner.
        3. The following are example sprint goals on an eCommerce application:
            1. Implement basic shopping cart functionality including add, remove, and update quantities.
            2. Develop the checkout process: pay for an order, pick shipping, order gift wrapping, etc.
        4. The sprint goal can be used for quick reporting to those outside the sprint. There are always stakeholders who want to know what the team is working on, but who do not need to hear about each product backlog item (user story) in detail.
    b. A sprint backlog
        1. The sprint backlog is the other output of sprint planning.
        2. A sprint backlog is a list of the product backlog items the team commits to delivering *plus* the list of tasks necessary to delivering those product backlog items.
        3. Each task on the sprint backlog is also usually estimated.

c. Team selects items from the product backlog they can commit to completing
d. Tasks are identified and each is estimated (1-16 hours)
e. Collaboratively, not done alone by the ScrumMaster
f. Example of sprint backlog

As a vacation planner, I want to see photos of the hotels.

Code the middle tier (8 hours)
Code the user interface (4)
Write test fixtures (4)
Code the foo class (6)
Update performance tests (4)

Daily Scrum

In Scrum, on each day of a sprint, the team holds a daily scrum meeting called the "daily scrum.”

- Meetings are typically held in the same location and at the same time each day **by standing up**.
- Ideally, a **daily scrum meeting** is held in the morning, as it helps set the context for the coming day's work.
- These scrum meetings are strictly time-boxed to **15 minutes**.
- This keeps the discussion brisk but relevant **(Ham and Eggs).**
- The daily scrum meeting is not used as a problem-solving or issue resolution meeting. Issues that are raised are taken offline and usually dealt with by the relevant subgroup immediately after the meeting.
- During the daily scrum, each team member answers the following three questions:

  - What did you do yesterday?
  - What will you do today?
  - Are there any impediments in your way?

If a programmer stands up and says, "Today, I will finish the data storage module," everyone knows that in tomorrow's meeting, he will say whether or not he finished. This has the wonderful effect of helping a team realize the significance of these commitments, and that their commitments are to one another, not to some far-off customer or salesman.

Any impediments that are raised in the scrum meeting become the ScrumMaster's responsibility to resolve as quickly as possible. Typical impediments are:

My ____ broke and I need a new one today.
I still haven't got the software I ordered a month ago.
I need help debugging a problem with ______.
I'm struggling to learn ______ and would like to pair with someone on it.
I can't get the vendor's tech support group to call me back.
Our new contractor can't start because no one is here to sign her contract.
I can't get the ____ group to give me any time and I need to meet with them.
The department VP has asked me to work on something else "for a day or two."
In cases where the ScrumMaster cannot remove these impediments directly himself (e.g., usually the more technical issues), he still takes responsibility for making sure someone on the team does quickly resolve the issue.

Sprint Review

In Scrum, each sprint is required to deliver a potentially shippable product increment. This means that at the end of each sprint, the team has produced a coded, tested and usable piece of software.

So at the end of each sprint, a sprint review meeting is held. During this meeting, the Scrum team shows what they accomplished during the sprint. Typically this takes the form of a demo of the new features.

- Team presents what it accomplished during the sprint
- Typically takes the form of a demo of new features or underlying architecture
- Informal
    - 2-hour prep time rule
    - No slides

- Whole team participates
- Invite the world

Sprint Retrospective

After the Sprint Review meeting took place the Scrum Team and the Scrum Master get together for the Sprint Retrospective. In this meeting all team members **reflect on the past sprint and check three things: what went well during the sprint, what didn't, and what improvements could be made in the next sprint.** The meeting should be time-boxed (e.g. 3 hours)..

a. The sprint retrospective is usually the last thing done in a sprint.
b. Many teams will do it immediately after the sprint review.
c. The entire team, including both the ScrumMaster and the product owner should participate.
d. You can schedule a scrum retrospective for up to an hour, which is usually quite sufficient
e. Typically 15–30 minutes
f. Using this approach each team member is asked to identify specific things that the team should:
    - Start doing
    - Stop doing
    - Continue doing

Scrum Artefacts

Product Backlog

The agile product backlog in Scrum is a prioritized features list, containing short descriptions of all functionality desired in the product. When applying Scrum, it's not necessary to start a project with a lengthy, upfront effort to document all requirements. Typically, a Scrum team and its product owner begin by writing down everything they can think of for agile backlog prioritization. This agile product backlog is almost always more than enough for a first sprint. The Scrum product backlog is then allowed to grow and change as more is learned about the product and its customers.

A typical Scrum backlog comprises the following different types of items:

1. Features

2. Bugs
3. Technical work
4. Knowledge acquisition

Conceptually, the team starts at the top of the prioritized Scrum backlog and draws a line after the lowest of the high-priority items they feel they can complete. In practice, it's not unusual to see a team select, for example, the top five items and then two items from lower on the list that are associated with the initial five.

1. The requirements
2. A list of all desired work on the project
3. Ideally expressed such that each item has value to the users or customers of the product
4. Prioritized by the product owner
5. Reprioritized at the start of each sprint

Product Backlog Example

Profiles

- As a site member, I want to describe myself on my own page in a semi-structured way. That is, I can fill in predefined fields, but also have room for a free-text field or two. (It would be nice to let this free text be HTML or similar.)
- As a site member, I can fill out an application to become a Practitioner.
- As a Practitioner, I want my profile page to include additional details about me (i.e., some of the
- answers to my Practitioner application).
- As a site member, I can fill out an application to become a Trainer.
- As a Trainer, I want my profile page to include additional details about me (i.e., some of the
- answers to my Trainer application).
- As a Practitioner or Trainer, when I provide content to the site I want a small graphic associated
- with the content indicating I'm a Practitioner or Trainer. (For example, Amazon's "Top 500
- Reviewers" approach.)
- As a trainer, I want my profile to list my upcoming classes and include a link to a detailed page
- about each.
- As a site member, I can view the profiles of other members.

- As a site member, I can search for profiles based on a few fields (class attended, location, name).
- As a site member, I can mark my profile as private in which case only my name will appear.
- As a site member, I can mark my email address as private even if the rest of my profile is not.
- As a site member, I can send an email to any member via a form.
- As a site administrator, I can read practicing and training applications and approve or reject
- them.
- As a site administrator, I can edit any site member profile.

News

- As a site visitor, I can read current news on the home page.
- As a site visitor, I can access old news that is no longer on the home page.
- As a site visitor, I can email news items to the editor. (Note: this could just be an email link to
- the editor.)
- As a site a site editor, I can set the following dates on a news item: Start Publishing Date, Old
- News Date, Stop Publishing Date. These dates refer to the date an item becomes visible on the site (perhaps next Monday), the date it stops appearing on the home page, and the date it is removed from the site (which may be never).

Sprint Backlog

a. The sprint backlog is a list of tasks identified by the Scrum team to be completed during the Scrum sprint.
b. During the sprint planning meeting, the team selects some number of product backlog items, usually in the form of user stories, and identifies the tasks necessary to complete each user story.
c. Most teams also estimate how many hours each task will take someone on the team to complete.
d. The sprint backlog is commonly maintained as a spreadsheet, but it is also possible to use your defect tracking system or any of a number of software products designed specifically for Scrum or agile.
e. An example of a sprint backlog in a spreadsheet looks like this:

User Story	Tasks	Day 1	Day 2	Day 3	Day 4	Day 5	...
As a member, I can read profiles of other members so that I can find someone to date.	Code the ...	8	4	8	0		
	Design the ...	16	12	10	4		
	Meet with Mary about ...	8	16	16	11		
	Design the UI	12	6	0	0		
	Automate tests ...	4	4	1	0		
	Code the other ...	8	8	8	8		
As a member, I can update my billing information.	Update security tests	6	6	4	0		
	Design a solution to ...	12	6	0	0		
	Write test plan	8	8	4	0		
	Automate tests ...	12	12	10	6		
	Code the ...	8	8	8	4		

Sprint Burndown

As a definition of this chart we can say that the Burndown chart displays the *remaining effort for a given period of time*.

When they track product development using the Burndown chart, teams can use a sprint Burndown chart and a release Burndown chart. This article concentrates on the sprint Burndown chart as it is used on daily basis.

Sprint Burndown Chart

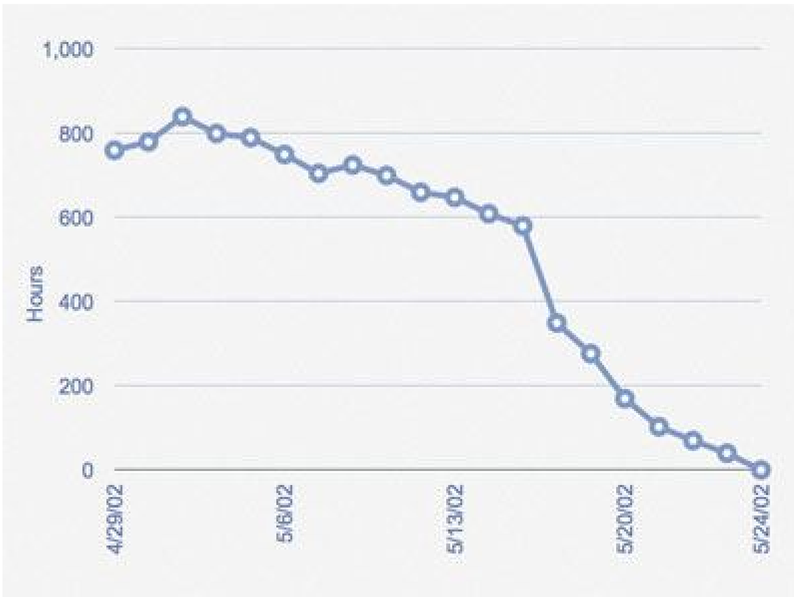
Teams use the sprint Burndown chart to track the product development *effort remaining in a sprint*.

General speaking the Burndown chart should consist of:

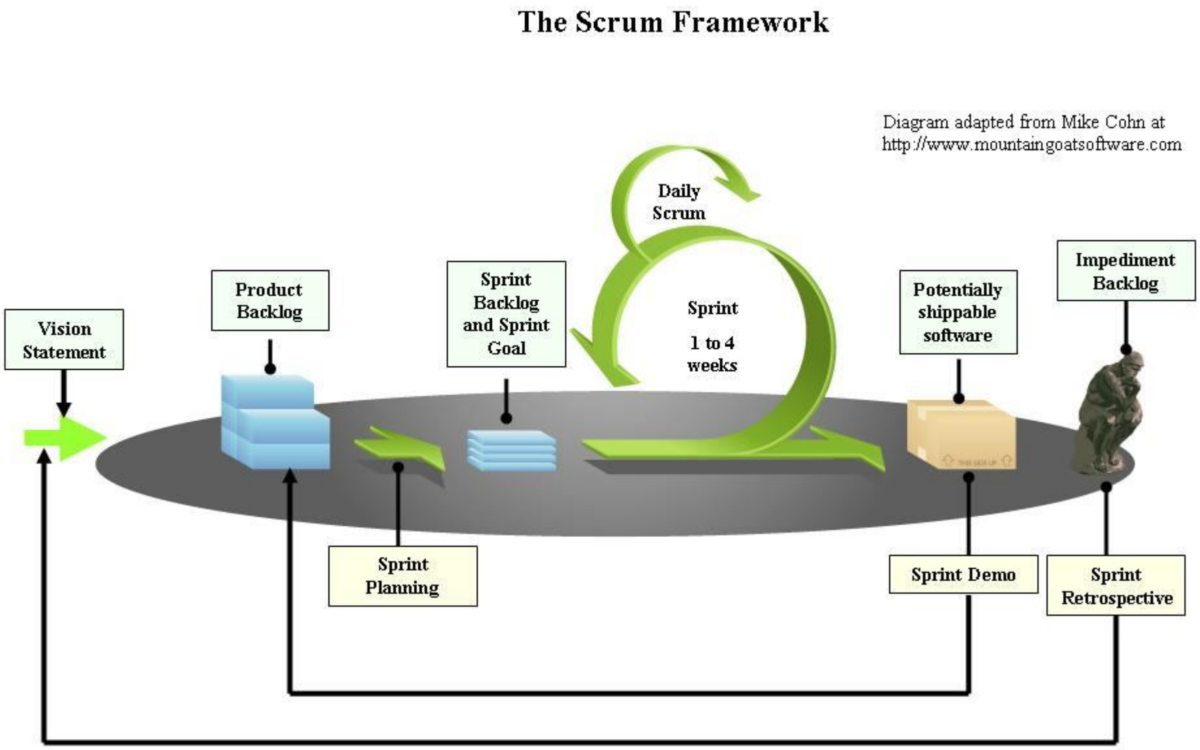
a. X axis to display working days
b. Y axis to display remaining effort
c. Ideal effort as a guideline
d. Real progress of effort

Companies use different attributes on the Y axis. All of them have benefits and drawbacks.

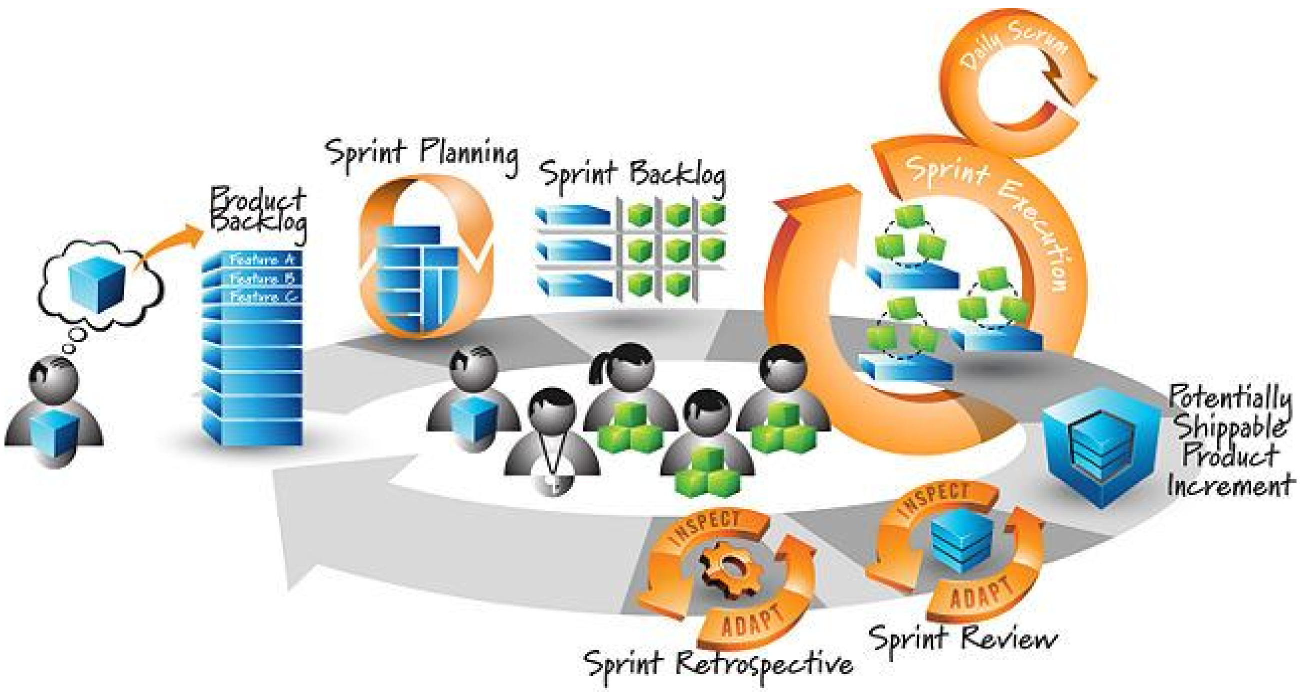
New agile teams used to start with the Burndown chart that displays the number of items on the Y axis.



Agile Scrum



By Md Saidul Haque, Sr. Software Engineer at Grameen Solutions



User stories

User stories are short, simple description of a feature told from the perspective of the person who desires the new capability, usually a user or customer of the system. They typically follow a simple template:

As a <type of user>, I want <some goal> so that <some reason>.

User stories are often written on index cards or sticky notes, stored in a shoe box, and arranged on walls or tables to facilitate planning and discussion. As such, they strongly shift the focus from writing about features to discussing them. In fact, these discussions are more important than whatever text is written.

One of the benefits of agile user stories is that they can be written at varying levels of detail. We can write a user story to cover large amounts of functionality. These large user stories are generally known as epics. Here is an epic agile user story example from a desktop backup product:

- As a user, I can backup my entire hard drive.

Because an epic is generally too large for an agile team to complete in one iteration, it is split into multiple smaller user stories before it is worked on. The epic above could be split into dozens (or possibly hundreds), including these two:

- As a power user, I can specify files or folders to backup based on file size, date created and date modified.
- As a user, I can indicate folders not to backup so that my backup drive isn't filled up with things I don't need saved.

What is Planning Poker?

Planning Poker is an agile estimating and planning technique that is consensus based. To start a poker planning session, the product owner or customer reads a agile user story or describes a feature to the estimators.

Each estimator is holding a deck of Planning Poker cards with values like 0, 1, 2, 3, 5, 8, 13, 20, 40 and 100, which is the sequence we recommend. The values represent the number of story points, ideal days, or other units in which the team estimates.

The estimators discuss the feature, asking questions of the product owner as needed. When the feature has been fully discussed, each estimator privately selects one card to represent his or her estimate. All cards are then revealed at the same time.

If all estimators selected the same value, that becomes the estimate. If not, the estimators discuss their estimates. The high and low estimators should especially share their reasons. After further discussion, each estimator reselects an estimate card, and all cards are again revealed at the same time.

The poker planning process is repeated until consensus is achieved or until the estimators decide that agile estimating and planning of a particular item needs to be deferred until additional information can be acquired.